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Research article

Gap tolerance and molten pool destabilization mechanism in oscillating laser-arc hybrid welding of aluminum alloys

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Outline

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Highlights

- An oscillating weld pool experiences periodic collapse before complete destabilization.
- A global force model predicts the critical gap needed to prevent pool destabilization based on weld seam geometry.
- Increasing the gap reduces the contact angle at the liquid-solid interface, weakening the vertical surface tension.
- In oscillating laser-arc hybrid welding, increasing the oscillation radius can improve the gap tolerance.

Abstract

Oscillating laser-arc hybrid welding (O-LAHW) offers significant advantages in enhancing efficiency and mechanical properties. However, in actual production, gap fluctuations can cause instability, limiting its application in large-scale structure manufacturing. In this study, we explored the impact of gap fluctuation on the destabilization of the molten pool in aluminum alloy O-LAHW and identified the maximum gap tolerance for various conditions. High-speed photography revealed that the oscillating molten pool undergoes a transitional state of periodic collapse before complete instability, a behavior distinct from that of a non-oscillating molten pool. We also analyzed the variations in weld geometry prior to destabilization and developed a global force model of the molten pool to identify key geometrical parameters and related driving forces contributing to destabilization. The results show that maintaining a surface tension ratio of over 55% at the root of the molten pool is crucial for its stability. Additionally, the effects of oscillatory behavior and gap variations on the laser-substrate interaction were explored, revealing the physical mechanisms behind the changes in key geometrical parameters of the melt pool cross-section. The centrifugal effect generated by high-frequency oscillation is identified as a crucial mechanism for extending the duration of periodic collapse compared to non-oscillating molten pools. By discussing the interactions between energy absorption, molten pool shape, and molten pool forces, the study reveals the evolution

4. Conclusions

CRedit authorship contribution statement

Declaration of Competing Interest

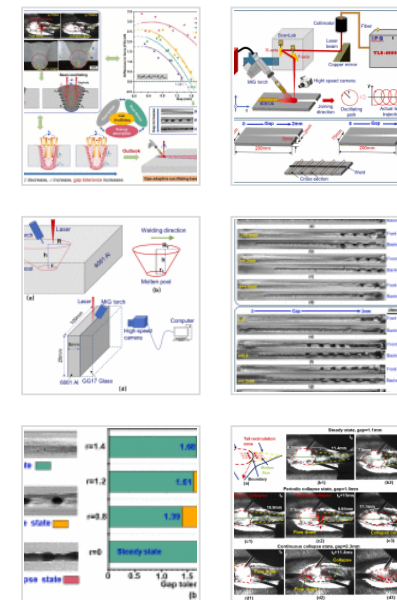
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Data availability

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Figures (15)



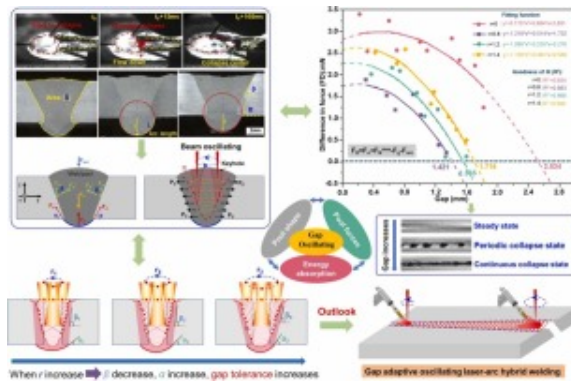
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Tables (1)

 Table 1. Welding process parameters.

process of weld destabilization and explains the differences in gap tolerance between oscillating and non-oscillating laser-arc hybrid welding, providing a reference for improving weld stability.

Graphical Abstract



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Keywords

Welding; Variable gap; Destabilization mechanism; Mechanical model; Force analysis

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1. Introduction

Laser-arc hybrid welding (LAHW) offers unparalleled advantages over [arc welding \[1\]](#) and [friction stir welding \[2\]](#) in terms of improving efficiency and mechanical properties.

However, in the welding of large components such as heavy machinery, high-speed rail

cars, and ultra-high voltage electrical switches, processing and assembly errors, as well as thermal deformation, can cause butt joint gap fluctuations of up to 3 mm or even higher [3]. These fluctuations in groove gaps can destabilize the LAHW molten pool, resulting in porosity [4] or even collapse [5], thus affecting the reliability and productivity of the welding process. Therefore, ensuring the stability of weld formation and quality under such complex industrial conditions is one of the key challenges currently faced by LAHW.

Due to the small range of laser interaction, the first to face the effect of gap fluctuation and receive research attention is in laser welding. For example, Zhang et al. [6] demonstrated that gap fluctuations cause instability in the absorption of laser energy by the workpiece and increase the complexity of molten pool flow. Chen et al. [7] found that the molten pool dynamics are highly sensitive to gap variations, and only when the gap is controlled within 0.4 mm can a continuous and stable weld be achieved. Although LAHW improves gap tolerance compared to laser welding due to the ability of the arc to increase the action width of the heat source and the supplementation of filler material, serious weld-forming defects can still occur in the face of large gap fluctuations. For instance, Turichin et al. [8] found that in 7 mm carbon steel butt joints, undercut appeared when the gap exceeded 0.6 mm, and the weld began to collapse when the gap exceeded 1.2 mm. Lamas et al. [9] showed that in LAHW of high-strength steel, when the gap reached 1.3 mm, the sidewalls of the weld root had less melting, poor wetting, and failed to form an effective liquid bridge, resulting in incomplete root fusion. Additionally, another of their studies revealed that as the gap increases linearly, the depression at the top of the weld gradually increases until eventual collapse occurs [10].

To suppress welding defects and improve gap tolerance, researchers have explored the causes of defects arising from gap variations. For example, Kim et al. [11] observed that an increase in gap alters the arc's conductivity and reduces heat transfer efficiency in LAHW, leading to abrupt changes in molten pool shape. Turichin et al. [8] also confirmed that as the gap increases, the efficiency of the hybrid heat source gradually decreases. In addition to directly affecting energy absorption and heat source efficiency, the gap increase reduces the workpiece's support force for the molten pool, causing molten pool to

collapse downward [5]. Moreover, Huang et al. [5] demonstrated that arc voltage influences the arc pressure on the molten pool, thereby altering its shape, and that adjusting the arc voltage can improve gap-bridging capability. Furthermore, it has been found that the surface tension at the weld root in LAHW is critical for maintaining molten pool stability [12], [13]. As we can see, existing studies have provided valuable insights into the effects of gap variations and molten pool stability, but the detailed mechanism of how the energy fluctuation, molten pool shape and molten pool force affect each other leading to welding instability is not clear. And most studies are mainly concentrated in high-strength steel and titanium alloy, aluminum alloy as the use of the second alloy after steel [14], the relevant research is less. Additionally, aluminum alloys have a lower surface tension coefficient [15], [16], leading to more intense molten pool flow, making aluminum welds more prone to depressions or even collapse.

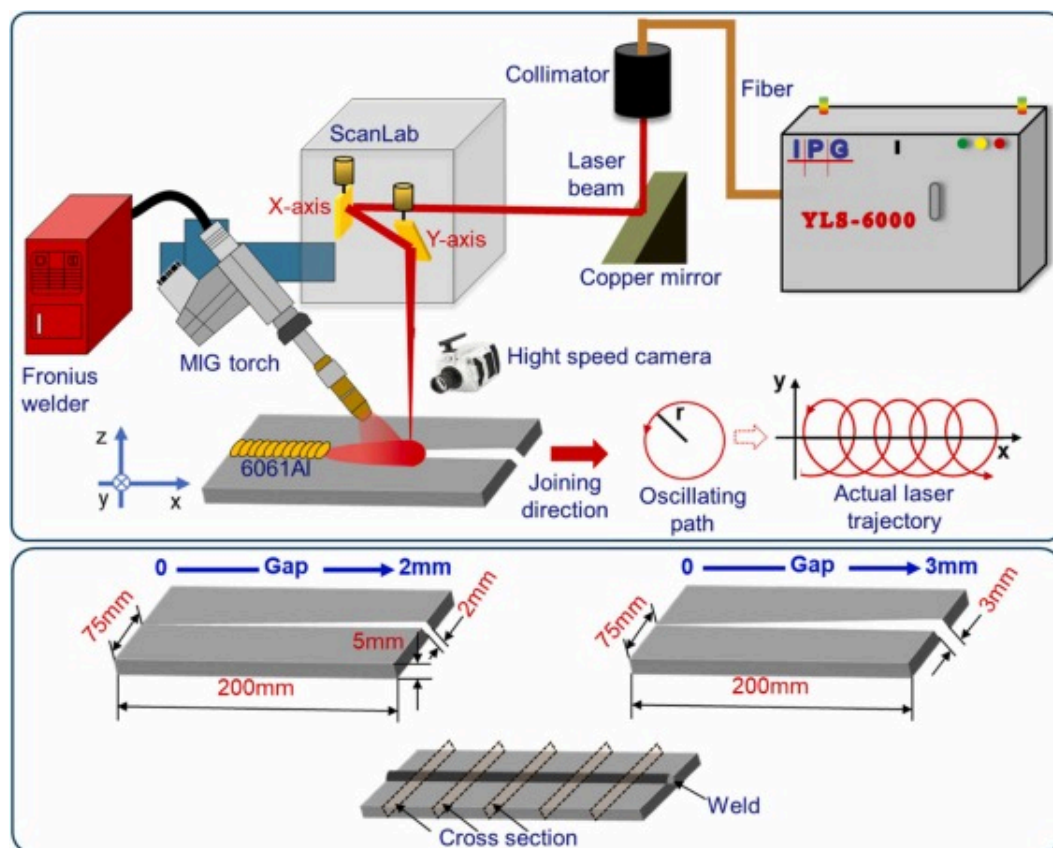
On the other hand, oscillating laser-arc hybrid welding, as a new technology, has significant advantages in addressing porosity issues and improving molten pool fluidity in aluminum alloy welding [17], [18], [19]. Moreover, beam oscillation has been proven to improve gap tolerance in laser welding by optimizing energy distribution [20], [21]. Whether introducing beam oscillation into laser-arc hybrid welding would further enhance gap tolerance remains unknown.

To this end, this study designed an experiment with varying gap sizes for aluminum alloy LAHW, comparing the gap tolerance of oscillating and non-oscillating welding. We established a mechanical model based on weld formation, quantifying for the first time the evolution of weld pool shape and forces under different parameters and gap conditions. By exploring the effects of gap variations and beam oscillations on the energy distribution as well as the absorption behavior of the base material, we have discovered the interplay law between energy absorption-melting pool shape-melting pool force and revealed the destabilization mechanism of aluminum alloys under large gaps. The findings of this study are of significant value for developing methods to improve gap tolerance in aluminum alloy LAHW and for promoting the broader application of this technology in engineering.

2. Experimental equipment and methods

2.1. Experimental equipment

The equipment used in the experiments included an IPG-YLS 6000 fiber laser, a hurrySCAN 30 oscillating galvanometer lens manufactured by SCANLAB, a Fronius welder, and a Fanuc robot, as shown in [Fig. 1](#). The laser wavelength is 1064 nm, the beam mode is TEM01, and the core diameter of the transmission fiber is 200 μm . The focal length of the oscillating weld head is 250 mm, and the beam diameter at the focal position is 0.52 mm. Since previous studies have shown that the circular oscillation mode strikes a good balance between energy distribution uniformity and penetration efficiency, achieving optimal weld shape, welding stability, and mechanical properties [\[22\]](#), [\[23\]](#), the circular oscillation mode will be used in this study. A Phantom V710 high-speed camera was used to capture the flow characteristics of the molten pool, with a frame rate of 10,000 frames per second and an exposure time of 1 μs . The wavelength of the background light source was 810 nm. ImageJ software was used to measure the weld cross-section data.



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Fig. 1. Schematic diagram of the experimental setup. The galvanometer controls the circular oscillation of the laser beam. The experimental equipment was described in more detail in a previous study [24]. Two gradient gaps, 0–2 mm and 0–3 mm, were preset in advance, and the cross-sectional morphologies at different positions were captured after welding.

The base material used for the experiments was 5 mm thick A6061 aluminum, with dimensions of 75 mm × 200 mm. Two gap variation rates were designed: one varying

asymptotically from 0 mm to 2 mm, and the other from 0 mm to 3 mm, as shown in Fig. 1.

The process parameters used for the experiment are presented in Table 1.

Table 1. Welding process parameters.

Test No.	Oscillation radius, r (mm)	Oscillation frequency, f (Hz)	Laser power, P (kW)	Average current, I (A)	Welding speed, v_w ($\text{m} \cdot \text{min}^{-1}$)	Gap (mm)
1	0	0	5.5	200	2	From 0 incremental to 2 mm
2	0.8	300	5.5	200	2	
3	1.2	300	5.5	200	2	
4	1.4	300	5.5	200	2	
5	0	0	5.5	200	2	From 0 incremental to 3 mm
6	0.8	300	5.5	200	2	
7	1.2	300	5.5	200	2	
8	1.4	300	5.5	200	2	

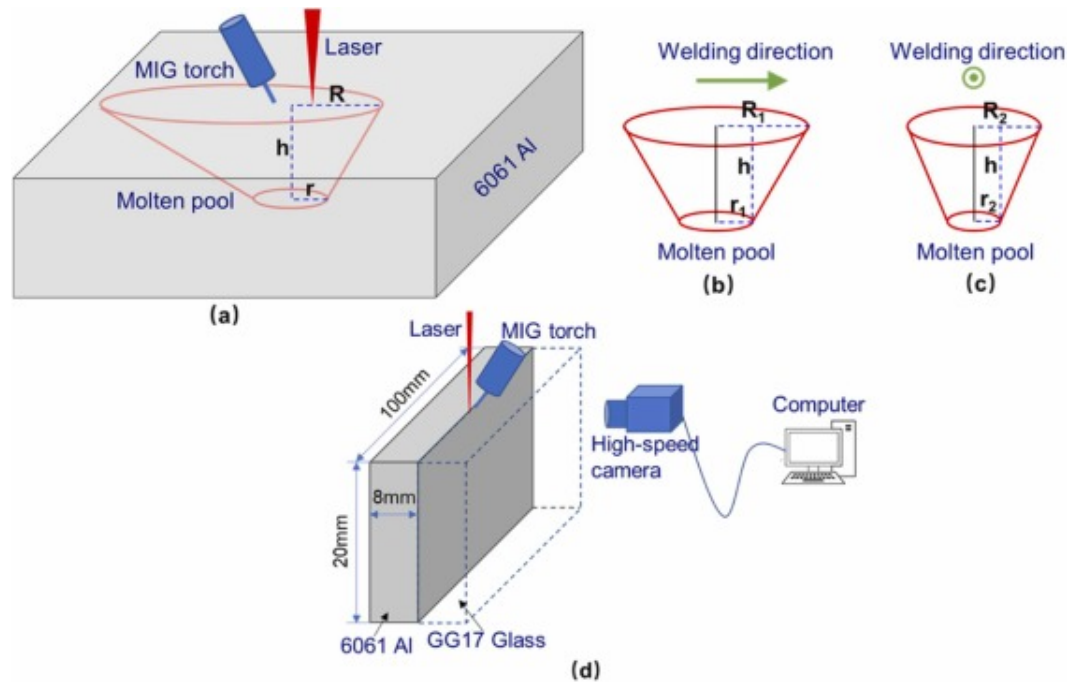
2.2. Molten pool volume calculation method

To simplify gravity calculations, the melt pool is approximated as a circular frustum, as shown in Fig. 2, with its volume calculated as follows:

$$V = \frac{1}{3}\pi h(r^2 + R^2 + rR) \quad (1)$$

where h is the depth of the molten pool, uniformly calculated for a base material plate thickness of 5 mm, and R and r are the radii of the upper and lower circles of the molten pool, respectively, as depicted in Fig. 2a. Since changes in the gap affect the melt pool width in the vertical welding direction, while the width along the welding direction is

primarily influenced by the oscillation parameters [6], [25], the width of the molten pool in both the welding direction (Fig. 2b) and the vertical welding direction (Fig. 2c) were measured separately. The radius of the upper circle of the cone was determined as $R = (R_1 + R_2)/2$, and the radius of the lower circle was determined as $r = (r_1 + r_2)/2$. To measure the values of R_1 and r_1 , the side morphology of the molten pool was captured using high-speed photography under different parameters, as referenced in [17], as shown in Fig. 2d.



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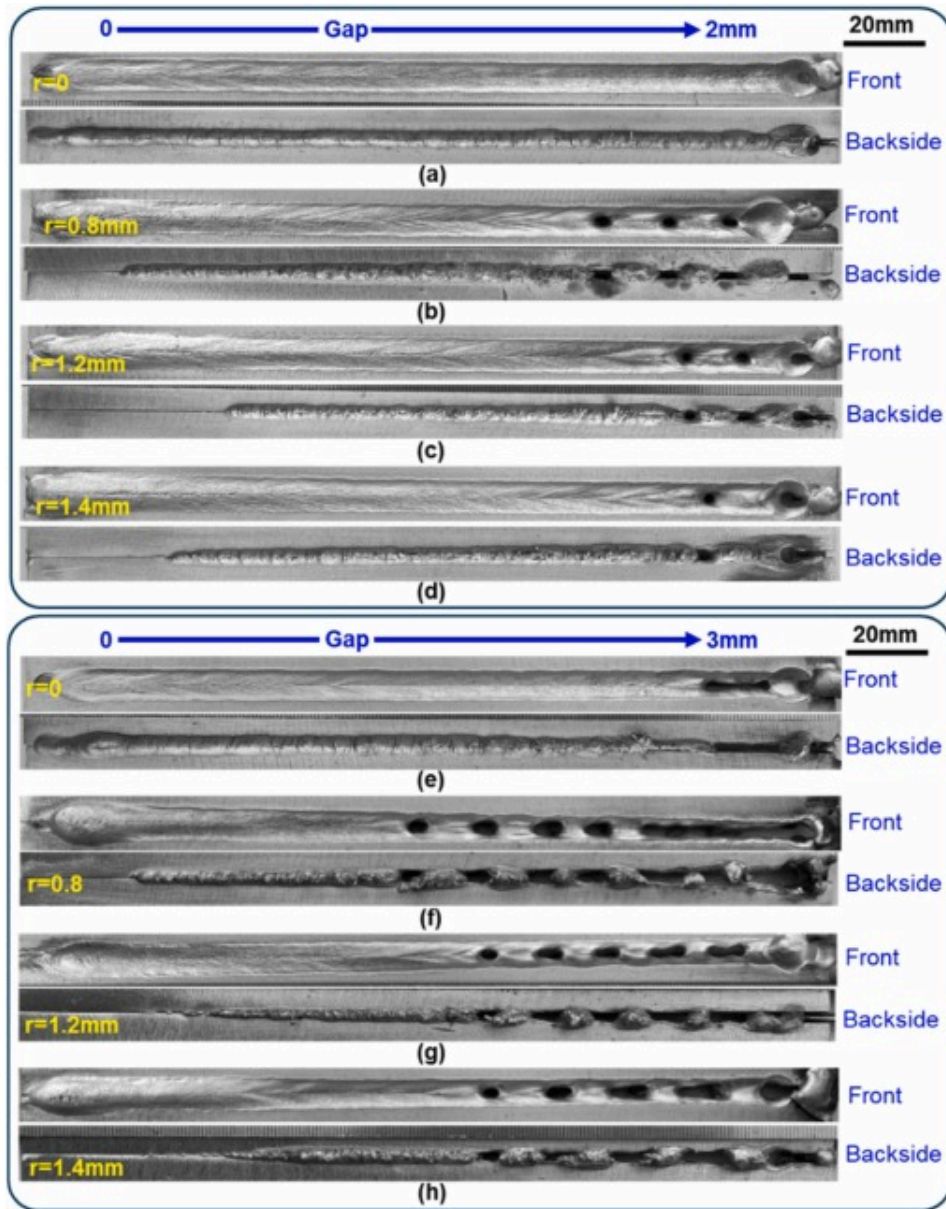
Fig. 2. Schematic diagram of the equivalent dimensions of the molten pool and measurement method. (a) The molten pool shape is approximated as an inclined circular frustum, with the actual pool dimensions being longer in the welding direction and shorter in the vertical welding direction, so the measurements were taken separately for each direction. (b) Schematic of the molten pool along the welding direction. (c)

Schematic of the molten pool in the vertical welding direction. (d) Method for measuring dimensions along the welding direction [17].

3. Results and discussion

3.1. Weld formation

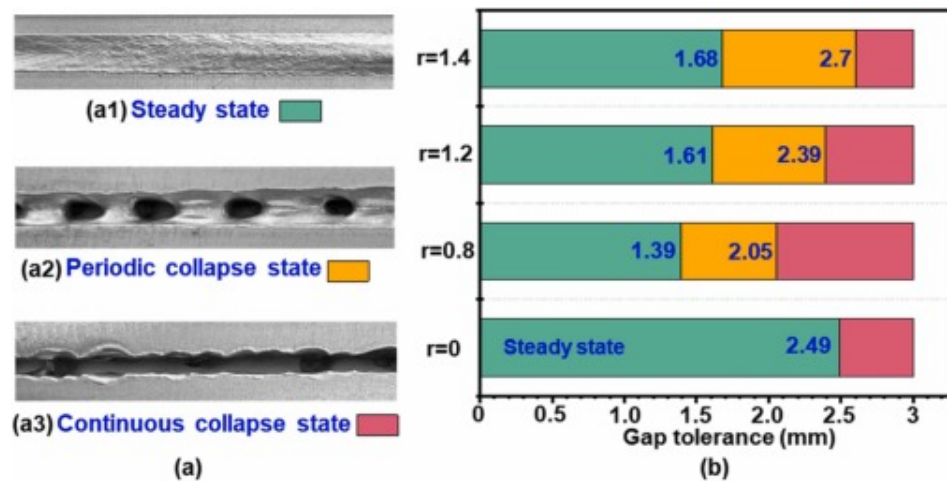
The weld shapes produced with different gaps and parameters are shown in Fig. 3. As the gap increases from 0 to 2 mm, the weld seam in LAHW remains stable with no significant collapse. However, with oscillation, collapse holes begin to appear, and this collapse becomes more pronounced as the gap increases to 3 mm, as shown in Fig. 3e,f. For destabilized welds, the weld progresses through three stages: the stabilization stage, followed by the periodic collapse stage, and finally the continuous collapse stage, as depicted in Fig. 4a. Based on the gap value at which the weld first shows signs of collapse, the gap tolerance for different parameters was calculated as shown in Fig. 4b. It was found that the gap tolerance in LAHW is 2.49 mm. When the oscillation radius increases from 0.8 mm to 1.4 mm, the gap tolerance increases from 1.39 mm to 1.68 mm, representing a 20.9 % improvement. However, continuous collapse of the weld in LAHW occurs very rapidly without an intermediate cyclic collapse phase, as shown in Figs. 3d and 4b.



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Fig. 3. Weld formation results. (a-d) Gap is incremented from 0 to 2 mm, (a) $r=0$, (b) $r=0.8$ mm, (c) $r=1.2$ mm, (d) $r=1.4$ mm. (e-h) Gap is incremented from 0 to 3 mm, (e) $r=0$, (f) $r=0.8$ mm, (g) $r=1.2$ mm, (h) $r=1.4$ mm.



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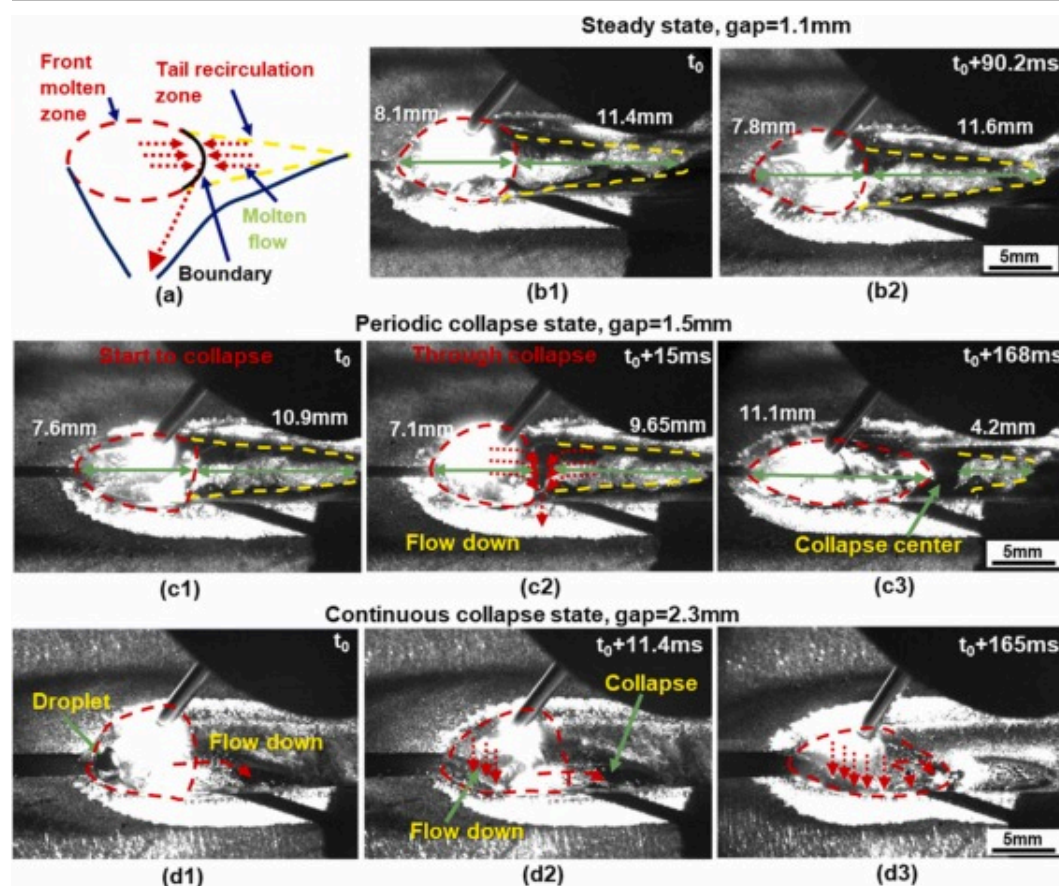
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Fig. 4. Statistical results of gap tolerance. (a) The three states of weld appearance. (b) Gap values correspond to different state transitions for various oscillation parameters, where different colors correspond to different stages of (a1) (a2) (a3).

3.2. Characteristics of molten pool instability

The flow characteristics of the molten pool at different gaps with fixed parameters were captured using high-speed photography, as shown in Fig. 5. For the convenience of discussion, the molten pool was divided into two regions: the melting zone at the front end (indicated by the red dashed circle in Fig. 5a) and the reflow zone at the tail end (indicated by the yellow area in Fig. 5a). At a gap of 1.1 mm, the weld is in a steady state, and the size and shape of the two zones remain essentially unchanged during welding, as shown in Fig. 5b1,b2. When the gap increases to 1.5 mm, the weld enters the periodic

collapse stage. Collapse initially occurs in the center of the front melting zone of the molten pool, as shown in [Fig. 5c1](#). As the front melting zone moves through the collapse region, the molten pool at the boundary between the melting zone and the reflow zone begins to flow downward, as shown in [Fig. 5c2](#). The length of the front melting zone decreases from 7.6 mm to 7.1 mm (indicated by the red dotted circle), and the tail reflow zone decreases from 10.9 mm to 9.65 mm. The two zones gradually start to separate. After passes through the collapse region, the front melting zone gradually increases to 11.1 mm, as shown in [Fig. 5c3](#). However, as the collapse hinders the melt flow exchange between the front melting zone and the tail reflow zone, the heat in the tail zone is not replenished, leading to solidification and a reduction to 4.2 mm, as indicated by the yellow dotted line in [Fig. 5c3](#). When the gap increases to 2.3 mm, the weld enters the continuous collapse stage. The reflow zone at the end of the molten pool essentially disappears, and the melt stream in the front melting zone flows obliquely downward along the tail. Due to the large gap, the front laser area of the molten pool fails to form, causing molten droplets to fall directly into the front gap area after transition, as shown in [Fig. 5d1](#). The falling droplets drive the melt pool in the front melting zone to flow downward, resulting in simultaneous downward flow of the melt stream in all directions. This causes irregular changes in the shape of the molten pool, preventing the maintenance of an effective melt pool, as indicated by the red dashed circle in [Fig. 5d](#).



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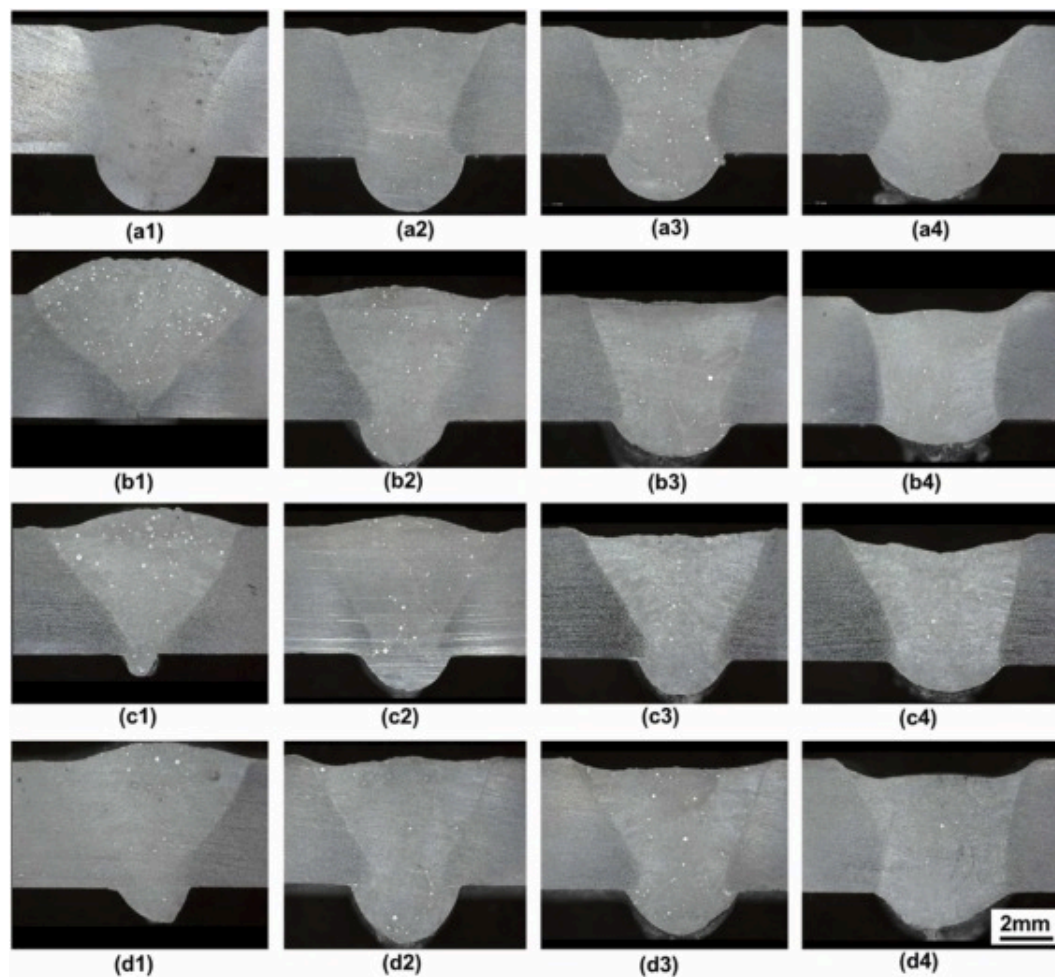
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Fig. 5. Melt pool flow characteristics at different gaps with an oscillating radius of 0.8 mm. (a) Schematic diagram of the melt pool, (b1-b2) the results at different moments of the steady state stage at a gap of 1.1 mm, (c1-c3) the results of the periodic collapse stage at a gap of 1.5 mm, (d1-d3) the results of the continuity collapse stage at a gap of 2.3 mm.

3.3. Evolution of weld cross-section shape

The cross-sectional morphology of the molten pool under various parameters and gaps is shown in Fig. 6. It can be observed that with the same parameters, the overall shape of

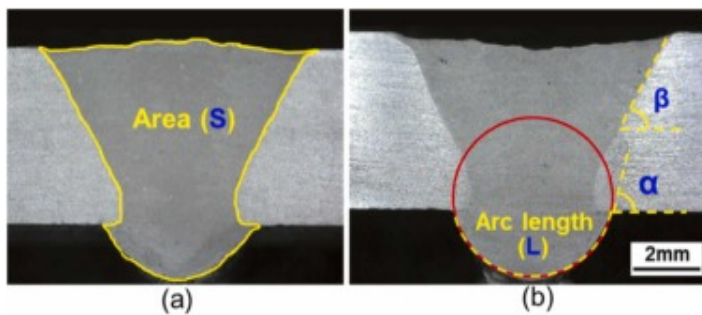
the weld cross-section gradually transitions from a V-shape to a U-shape as the gap increases. This transformation is particularly evident when beam oscillation is introduced, as shown in Fig. 6b-d. To further clarify the evolution of the weld cross-section under different parameters and gaps, the area (S), root arc length (L), and critical angles (α and β) of the weld cross-section were measured, as illustrated in Fig. 7.



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Fig. 6. Cross-sectional morphology of welds under various oscillation parameters and gaps. (a1-a4) Non-oscillating weld sections, (a1) gap=0.22 mm, (a2) gap=0.60 mm, (a3) gap=1.0 mm, (a4) gap=1.41 mm. (b1-b4) Weld cross-section at an oscillation radius of 0.8 mm, (b1) gap=0.30 mm, (b2) gap=0.68 mm, (b3) gap=1.01 mm, (b4) gap=1.57 mm. (c1-c4) Weld cross-section at an oscillation radius of 1.2 mm, (c1) gap=0.37 mm, (c2) gap=0.64 mm, (c3) gap=1.06 mm, (c4) gap=1.33 mm. (d1-d4) Weld cross-section at an oscillation radius of 1.4 mm, (d1) gap=0.42 mm, (d2) gap=0.81 mm, (d3) gap=1.14 mm, (d4) gap=1.60 mm.



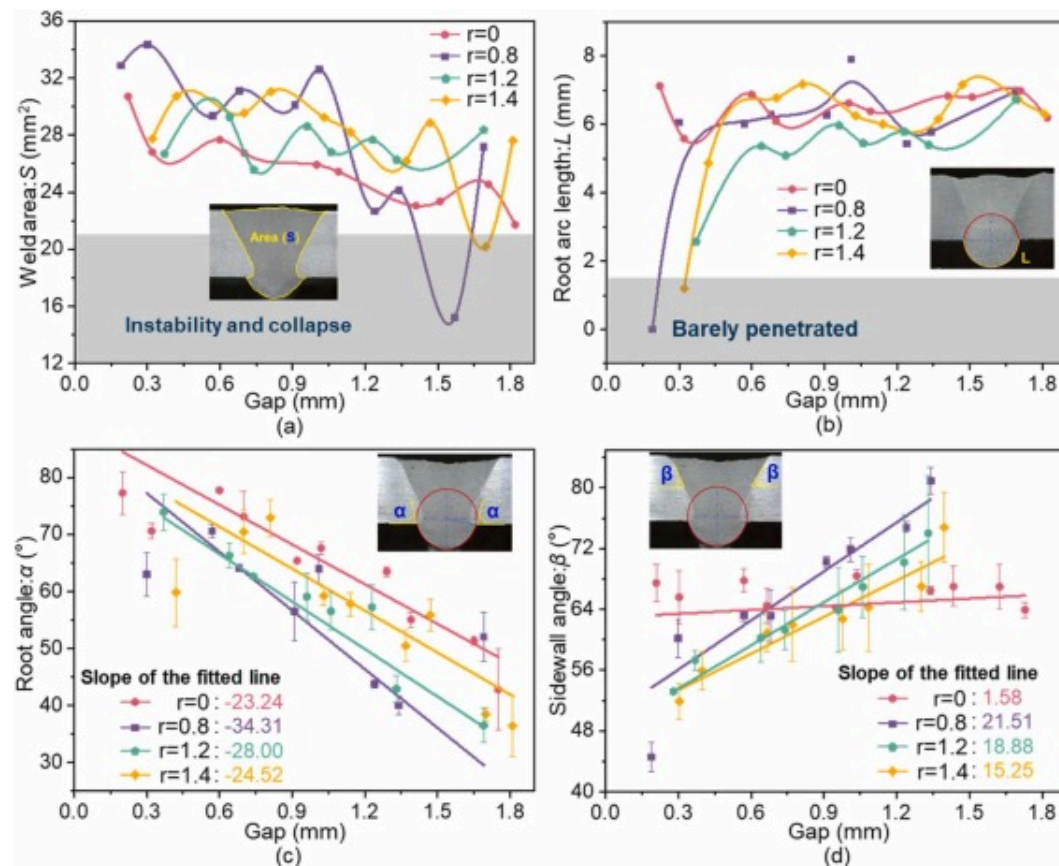
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Fig. 7. Key geometrical parameters of weld cross-section. (a) Area of the melt pool. (b) Arc length of the melt pool root, tangential angle α of the liquid-solid interface at the root, and angle β at the beveled edge of the solid-liquid interface along the side of the melt pool.

The weld cross-section area decreases gradually with an increasing gap is shown in Fig. 8a. Generally, the weld area is smaller when there is no oscillation, as indicated by the red curve in Fig. 8a. Fig. 8b presents the results for the root arc length, which remains relatively stable around 6 mm, except in cases where the weld is not fused. There is no significant variation between different parameters in this regard. From Fig. 8c, it is clear that the tangential angle (α) (as defined in Fig. 7b) gradually decreases as the gap

increases, with α being the largest for the same gap when there is no oscillation. Linear fitting reveals that the slope of the fitted line decreases from 34.31 to 24.52 as the oscillation radius increases from 0.8 mm to 1.4 mm. In contrast, the slope is only 23.24 without oscillation, indicating the slowest rate of monotonic decrease. Fig. 8d shows the variation in the ramp angle (β) (as defined in Fig. 7b), which increases gradually with the gap. The linear fit results indicate that the growth rate of β slows as the oscillation radius increases. Without oscillation, the slope of the fit is only 1.58, indicating the slowest growth rate.



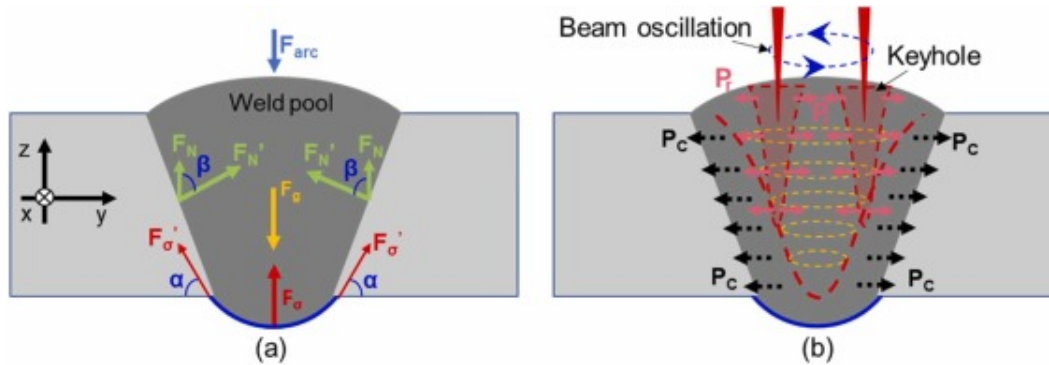
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Fig. 8. Statistics analysis of key geometrical parameters of the weld cross-section with different parameters and gaps. (a) Variation in weld area (S), (b) variation in root arc length (L), (c) variation in tangential angle (α), (d) variation in beveled angle (β). Each data point in (c) and (d) represents the average of the measurements from the weld cross-sections on both sides, with the error bars indicating the range between the maximum and minimum values.

3.4. Analysis and modeling of molten pool forces

The forces acting during the welding process are crucial in determining the internal flow of the molten pool and the resulting weld shape [25]. To explore the internal destabilization mechanism of the molten pool, a force model was developed based on weld shaping and beam oscillation. The molten pool, as a whole, is primarily influenced by gravity, arc pressure, surface tension, droplet impact force, and the support force of the base material [5], [26]. Given the low density of aluminum alloy and the small mass of the molten droplet, roughly 0.03 mN [27], the droplet impact force is minimal and can be considered negligible in the context of the overall molten pool dynamics. Additionally, the metal vapor reaction force and laser impact force form an equilibrium within the keyhole [28], allowing them to be disregarded when considering the molten pool as a whole. Thus, the overall force model can be simplified, as shown in Fig. 9a. To balance the horizontal metal vapor pressure and hydrostatic pressure within the laser keyhole, a pressure (P_r) is exerted along the vertical laser optical axis [26]. When the beam oscillates at high frequency, this pressure (P_r) generates a centrifugal effect on the molten pool, resulting in the formation of virtual centrifugal pressure (P_c), as illustrated in Fig. 9b.



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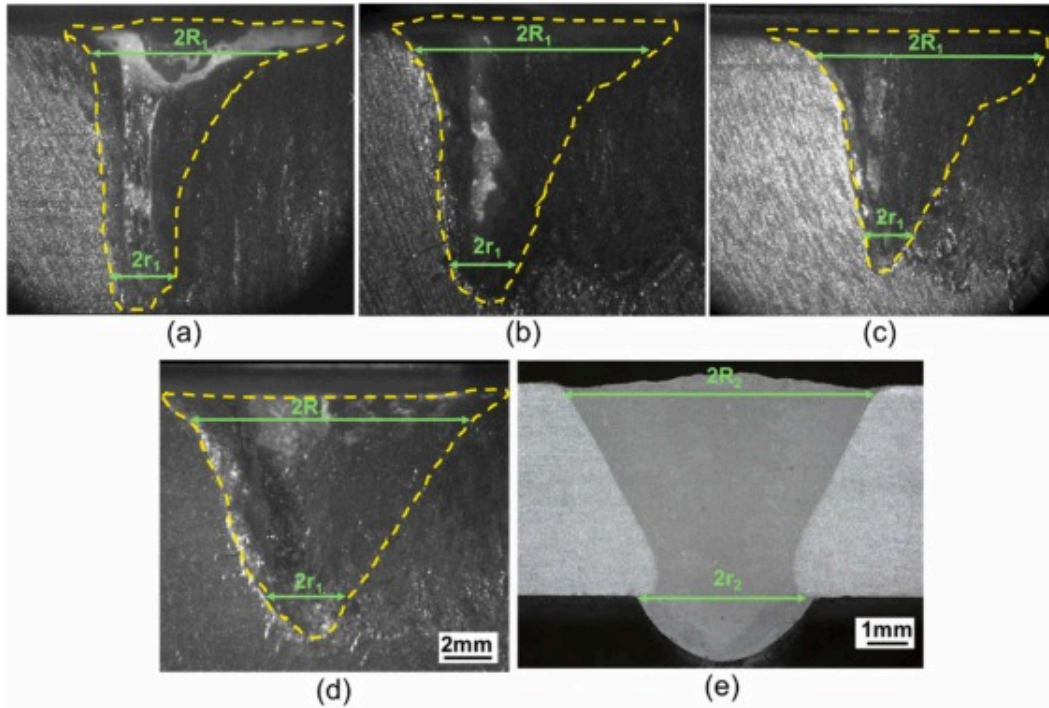
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Fig. 9. **Schematic diagram of the global force model.** (a) Force model of the molten pool, where F_N' represents the support force of the inclined plane on the molten pool, F_N is the vertical component of this support force, F_{σ}' is the surface tension at the root of the molten pool, and F_{σ} is the vertical component of this surface tension. (b) Virtual centrifugal pressure (P_C) generated by the molten pool under high-speed oscillation.

The gravity (F_g) acting on the melt pool is calculated using the following equation:

$$F_g = mg = \rho_L Vg \quad (2)$$

where m represents the mass of the melt pool, determined from the density of liquid aluminum alloy ($\rho_L=2.8 \text{ g/cm}^3$) [29] and the melt pool volume (V). The gravitational acceleration (g) is taken as 9.8 kg/N . The volume of the molten pool was determined using the method described in subsection 2.2. Additionally, the side profile of the molten pool was captured using a high-speed camera under different oscillation parameters, as shown in Fig. 10. The upper width of the molten pool was measured as R_1 , and the lower width as r_1 . The cross-sectional dimensions of the molten pool in the vertical welding direction were measured, as illustrated in Fig. 10e. These measurements were recorded for various parameters and gap sizes, and the volume was calculated using Eq. 1.



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Fig. 10. Measurement of molten pool dimensions in different directions. (a-d) Cross-sectional dimensions of the molten pool along the welding direction for various parameters: (a) $r = 0$, (b) $r = 0.8$ mm, (c) $r = 1.2$ mm, (d) $r = 1.4$ mm. (e) Cross-sectional dimensions of the molten pool in the vertical welding direction.

The arc pressure (F_{arc}) applied to the upper surface of the molten pool is composed of two main components: the plasma flow force (F_p) and the axial electromagnetic force (F_{ema}) [5], [28]. The plasma flow force can be calculated using Eq. 3 [21]:

$$F_p = \frac{1}{2} C_p \pi r_d \rho_p (KI)^2 \quad (3)$$

where C_p is the plasma flow force coefficient ($C_p=0.45$), r_d is the radius of the melt droplet ($r_d=0.65$ mm), ρ_p is the arc plasma density ($\rho_p=5 \times 10^{-2}$ kg/m³), K is the scaling factor ($K=7$)

[27], and I is the arc current. The axial electromagnetic force is calculated using Eq. 4 [30]:

$$F_{ema} = \frac{\mu_m I^2}{4\pi} \exp\left(\frac{-r_a^2}{2\sigma_j^2}\right) \quad (4)$$

where μ_m is the magnetic permeability ($4\pi \times 10^{-7}$ N/A²), σ_j is the current distribution coefficient ($\sigma_j = 2.53$ mm) [27], and r_a is the radius of the arc column ($r_a = 2.53$ mm) [29]. The total arc pressure (F_{arc}) is the sum of the plasma flow force and the axial electromagnetic force ($F_{arc} = F_p + F_{ema}$). As shown in (3), (4), the magnitude of F_{arc} is primarily influenced by the arc current [31].

Moreover, there is an interfacial bonding surface tension between the bottom of the melt pool and the solid base material [13], [29]. This surface tension can be decomposed into vertical and horizontal components, with the horizontal component being negligible due to mutual cancellation [32]. The formula for the surface tension (F_σ) in the vertical direction is given by Eq. 5 [29]:

$$F_\sigma = 2F'_\sigma \sin \alpha = L\sigma \sin \alpha \quad (5)$$

where L represents the length of the root arc, and the tangential angle (α) is determined by the tangent's position on the arc, as shown in Fig. 7b. The statistics values for L and α are presented in Fig. 8b,c, respectively. The surface tension coefficient (σ) is primarily influenced by temperature. Assuming a uniform melting temperature at the melt boundary, the surface tension coefficient of the aluminum alloy is 0.88 N/m [29].

In addition to the surface tension supporting the molten pool, the solid-liquid interface at both side of the molten pool also exerts a support force (F_N) on the pool. This force is primarily generated by the reaction force of gravity and arc pressure acting on the inclined surface. According to Newton's second law and ignoring the effects of friction, the formula for the support force is given by Eq. 6:

$$F_N^{\max} = (F_g + F_{arc}) \cos^2 \beta \quad (6)$$

where β is the angle of the beveled edge of the solid-liquid interface at both side of the molten pool, as shown in Fig. 7b. It is important to note that the F_N^{max} calculated here represents the maximum support force that can be provided at different gaps. When the melt pool is in equilibrium, the actual support force is less than this maximum value. Therefore, the discussion of support force in the following sections will focus on this maximum limit value.

According to the laws of circular motion, the virtual centrifugal pressure (P_c) in an oscillating molten pool can be calculated using Eq. 7:

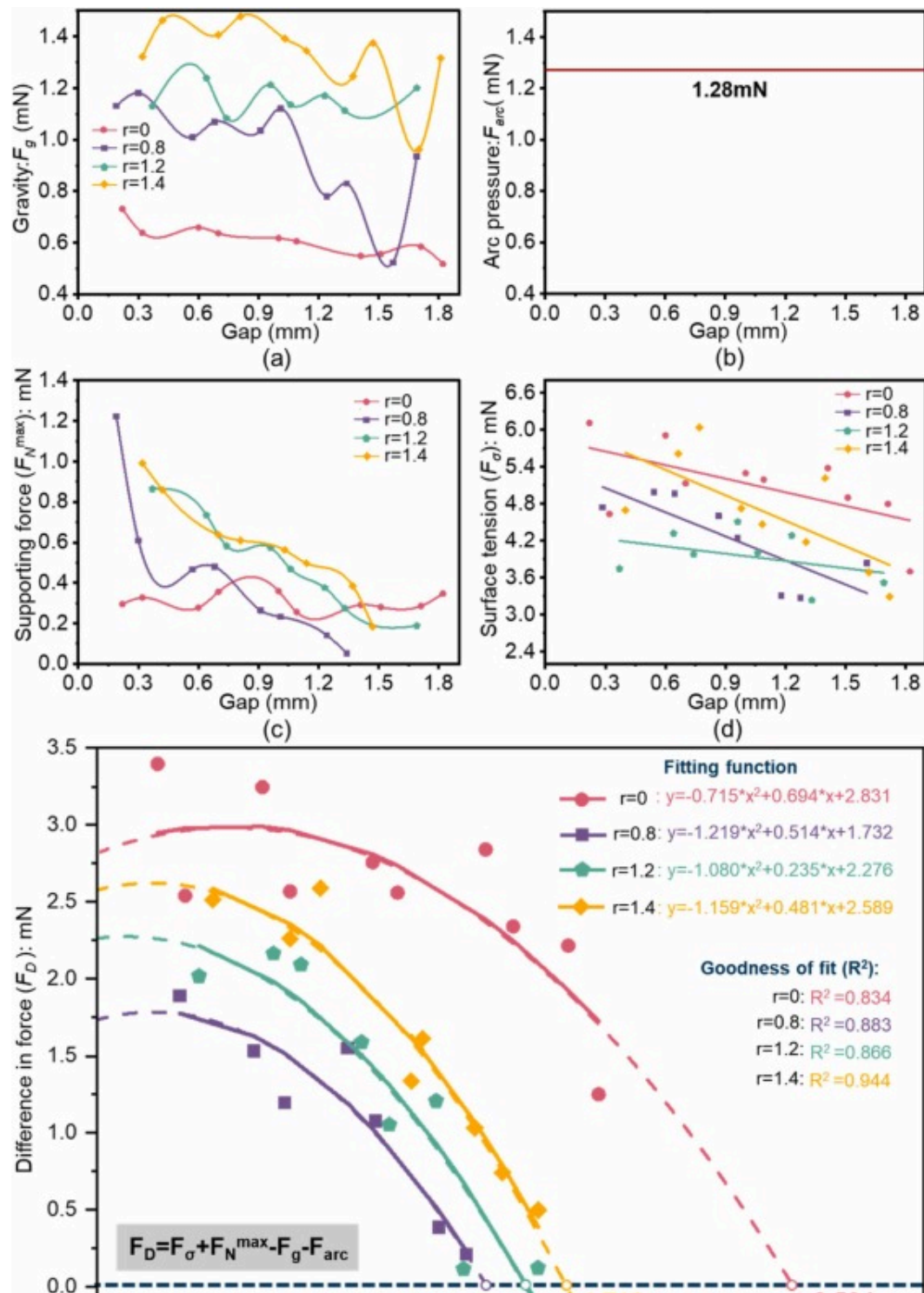
$$P_C = \frac{\rho v^2}{2} = 2\pi^2 \rho_L f^2 r^2 \quad (7)$$

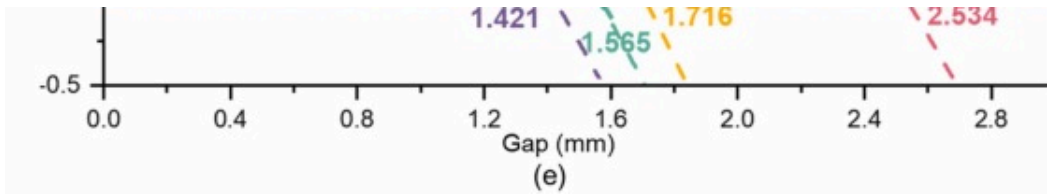
where ρ_L is the density of the liquid aluminum alloy molten pool ($\rho_L = 2.8 \text{ g/cm}^3$), and f and r represent the frequency and radius of oscillation, respectively.

3.5. Evolution of the molten pool forces

Based on the force model and the statistical results of the weld cross-section geometrical parameters, the variation of the force in the vertical direction with different parameters was calculated, as shown in Fig. 11. From Fig. 11a, it can be observed that the gravity decreasing as the gap increases. The gravity on the melt pool is at its minimum without oscillation at the same gap (as shown by the red curve in Fig. 11a) and increases progressively with the oscillation radius. Compared to LAHW, the gravity of the O-LAHW molten pool increases by over 65 % due to the larger melt pool width and volume resulting from the increased oscillation radius. Since the arc current is kept constant in the experiment, the arc pressure remains constant across different parameters. The vertical arc pressure (F_{arc}) of 1.28 mN is obtained by substituting numerical calculations according to (3), (4) shown in Fig. 11b. The results in Fig. 11c show that the support force fluctuates within a certain range in LAHW (as indicated by the red line in Fig. 11c), and gradually decreases with increasing gap in O-LAHW. Linear fitting results indicate that the vertical surface tension decreases progressively with the increase in gap size, as shown in Fig. 11d. At the same gap, the surface tension of the non-oscillating molten pool

is the largest (as indicated by the red curve in [Fig. 11d](#)), decreasing by 10.9–24.5 % after oscillation.





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Fig. 11. Variation of forces in the melt pool with different parameters and gaps. (a) The variation of gravity (F_g), (b) the calculation of arc pressure (F_{arc}), (c) the variation of support force (F_N^{max}), (d) the variation of surface tension (F_σ), (e) the results of the fitting of the difference in force (F_D) in the vertical direction, which is equal to the vertically upward force minus vertically downward force, and the dotted line in the graph is the extension of the quadratic fitting function.

The difference between the vertical upward force and the vertical downward force (F_D) is calculated using Eq. 8, with the results presented in Fig. 11e:

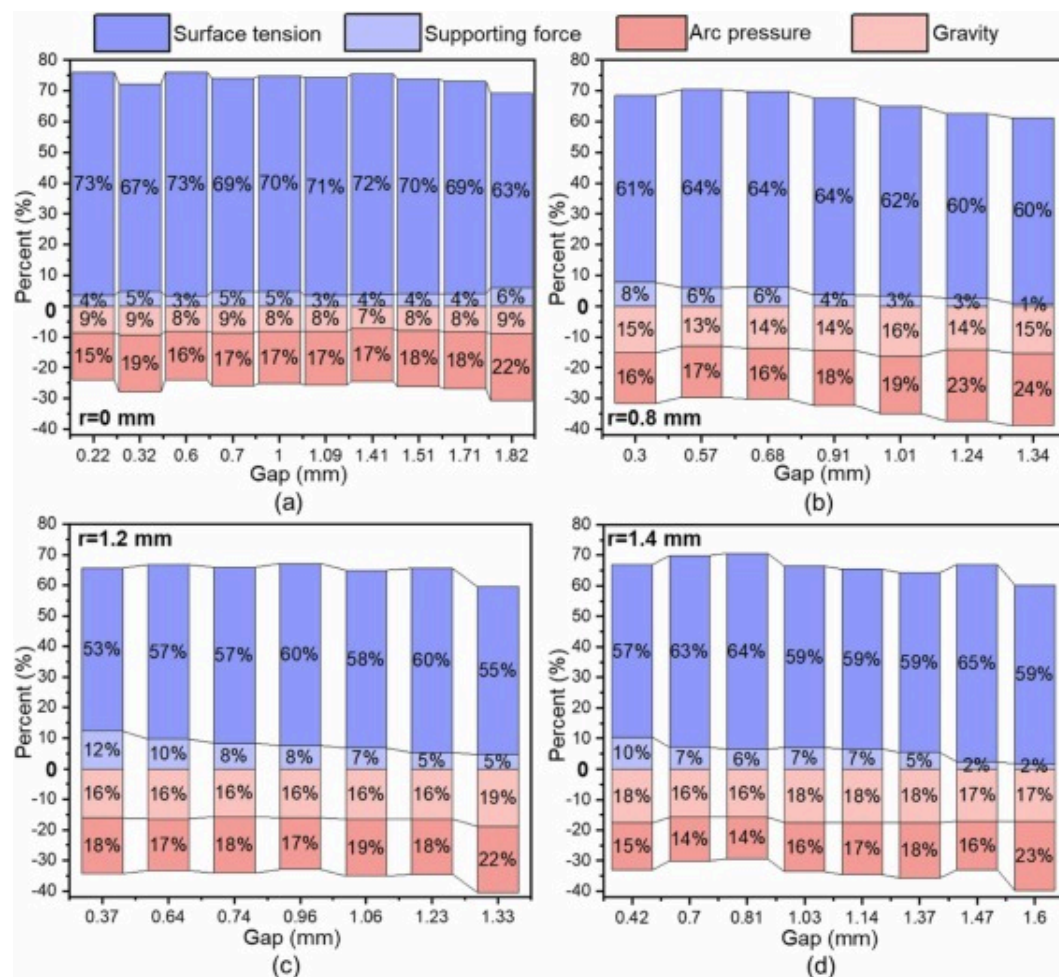
$$F_D = F_\sigma + F_N^{max} - F_g - F_{arc} \quad (8)$$

The stability of the molten pool is determined by the competition between these downward and upward forces. When the downward resultant force is less than the upward resultant force (i.e., F_D is greater than 0), the melt pool remains in a steady state, as shown in Fig. 4a1. When the downward resultant force equals the upward resultant force (i.e., F_D is equal to 0), the molten pool enters a periodic collapse stage, as shown in Fig. 4a2. Finally, when the downward resultant force exceeds the upward resultant force (i.e., F_D is less than 0), the melt pool transitions into a continuous collapse stage, as depicted in Fig. 4a3.

From the results in Fig. 11e, it can be observed that the F_D decreases slowly at first and then more rapidly as the gap increases. When the oscillating radius is 0, the relationship between the F_D and the gap can be fitted with the quadratic function:

$y = -0.715x^2 + 0.694x + 2.831$, with R^2 value of 0.834, indicating a good fit. The intersection point of this function with the X-axis is (2.534,0), which represents the gap threshold for the onset of molten pool instability at 2.534 mm when there is no oscillation. When the oscillating radius is 0.8 mm, the fitting function of the F_D with the gap change is $y = -1.219x^2 + 0.514x + 1.732$, with R^2 equal to 0.883. The threshold for destabilization is 1.421 mm. For oscillating radii of 1.2 mm and 1.4 mm, the instability gap thresholds predicted by the fitting functions are 1.565 mm and 1.716 mm, respectively. This indicates that sensitivity to instability is lowest in non-oscillating LAHW. The gap threshold of instability increases gradually with the increase of the oscillation radius in O-LAHW.

To reveal the differences in force evolution during the melt pool destabilization process, the percentage contributions of the four forces across different parameters as the gap changes were statistically analyzed, and the results are presented in Fig. 12. The vertical axis above the zero scale represents upward forces, while below it represents downward forces. It can be observed that surface tension consistently accounts for more than 50 % of the total resultant force for any parameter, making the largest contribution to the stability of the molten pool. Additionally, the percentage changes among the different forces reveal that the reduction in gap tolerance after oscillation is mainly due to two factors: firstly, oscillation increases the gravity of the molten pool, with the percentage of gravity after oscillation ranging from 13 % to 19 %, compared to less than 10 % without oscillation. Secondly, oscillation decreases the surface tension at the root of the molten pool, where the percentage of surface tension after oscillation is around 60 %, compared to approximately 70 % without oscillation. Furthermore, the percentage of vertical upward resultant force in LAHW is maintained near 70 % before collapse, as shown in Fig. 12a, while the percentage of resultant force in O-LAHW is maintained near 60 %. Previous discussions indicate that collapse begins when the upward resultant force share decreases to 50 %. This suggests that the transition from a steady state to continuous collapse occurs more rapidly in LAHW, while the occurrence of periodic collapse phases in O-LAHW can delay this process.



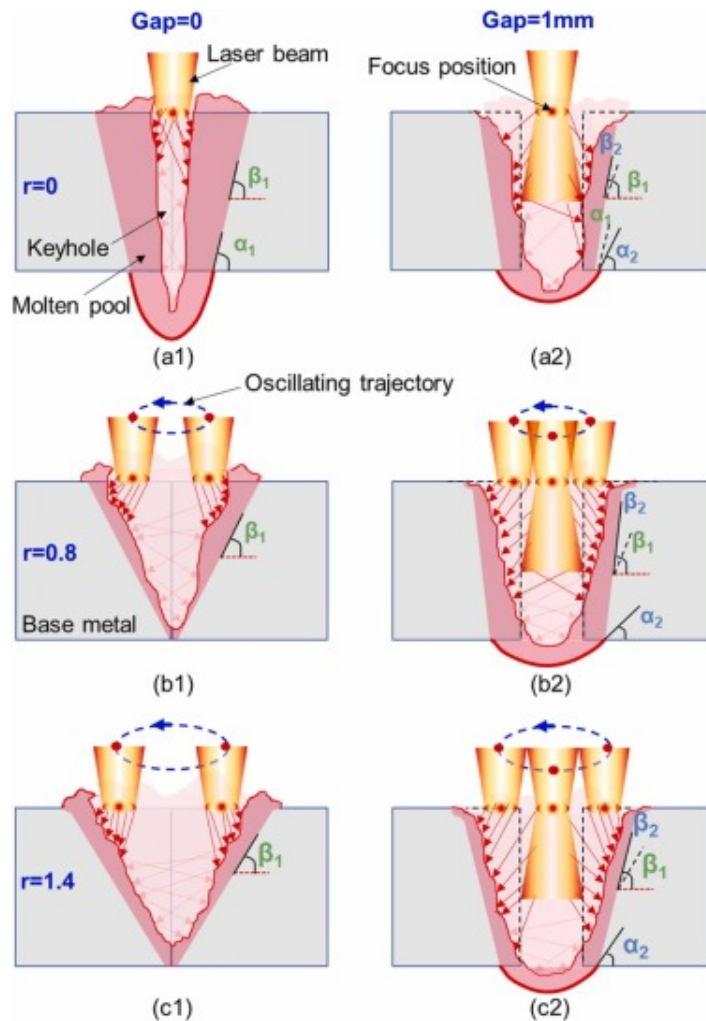
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Fig. 12. Percentage distribution of the four forces across different parameters and gap values. (a) $r=0$, (b) $r=0.8$ mm, (c) $r=1.2$ mm, (d) $r=1.4$ mm. The horizontal axis represents the gap values, while the vertical axis shows the percentage contributions of each force. Forces below the 0 scale, indicated by a negative sign, represent downward forces, including gravity and arc pressure.

3.6. Melt pool destabilization physics mechanisms

Since the forces within the molten pool are related to the weld shape, which is influenced by both process parameters [32], [33] and gap size [5], and given that the laser is a key factor affecting the molten pool shape in LAHW [33], it is crucial to understand these relationships. To detail the process of molten pool destabilization, Fig. 13 illustrates the interactions between energy absorption, molten pool shape, and molten pool forces under varying gaps and beam oscillation.



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Fig. 13. Physical mechanisms of melt pool shape change with increasing gap for different parameters. (a1,a2) $r=0$, (b1,b2) $r=0.8$ mm, (c1,c2) $r=1.4$ mm, (a1, b1, c1) gap=0, (a2, b2, c2) gap=1 mm. Red arrow represents the laser transmission path direction.

Firstly, in non-oscillating LAHW, when the gap is zero, most laser energy is absorbed by the base material at the focal plane, with additional absorption from Fresnel absorption after multiple reflections from the keyhole wall [34], [35]. This efficient energy absorption increases the molten pool's melting area and deposits more liquid metal at the root, raising the tangential angle (α_1) of the liquid-solid interface, as shown in Fig. 13a1. When the gap increases to 1 mm, the direct interaction area between the laser focal plane and the base material decreases, reducing absorption. Simultaneously, the larger gap facilitates the propagation of laser energy along the depth direction, widening the keyhole [36], as illustrated in Fig. 13a2. The metal vapor inside the keyhole causes Mie and Rayleigh scattering of the reflected laser, intensifying the plasma shielding effect [37], [38], further lowering absorption. The combined effect of a larger keyhole and decreased melting of the base material increases the angle (β_2) between the sidewall of the molten pool and the horizontal plane, as shown in Fig. 13a2. As the laser energy propagates downwards, it increases the melt width at the bottom of the pool, but the overall melting of the base material decreases, causing less liquid metal to flow back to the root of the molten pool and reducing the residual height [39]. This leads to a decrease in the tangential angle (α_2) at the liquid-solid interface at the root of the molten pool. Analytical calculations in 3.4 Analysis and modeling of molten pool forces, 3.5 Evolution of the molten pool forces show that increasing β reduces the molten pool's support force (F_N^{max}), while decreasing α lowers the vertical upward surface tension (F_σ).

The introduction of oscillation increases the melt pool size, with larger oscillation radii leading to larger pools [22], [26]. When the oscillation radius is 0.8 mm or 1.4 mm, the corresponding melt pool morphologies are shown in Fig. 13b1,b2,c1,c2. Oscillation enhances the contact area between the focal plane and the base material, widening the

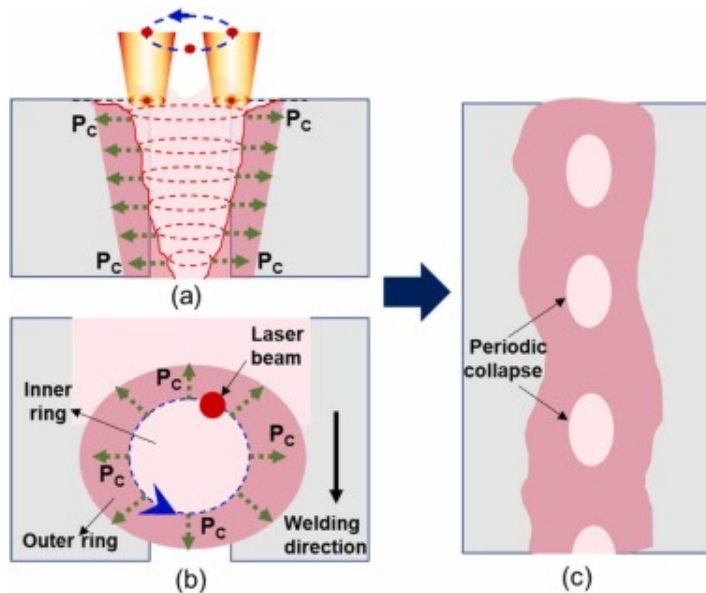
melt pool's top. A small portion of the laser beam reflects from the keyhole wall, creating a V-shaped pool with a shallower sidewall angle (β) compared to non-oscillating conditions, as shown in Fig. 13b1,c1. This results in double the support force for the oscillating pool when the gap is less than 0.5 mm (Fig. 12), while the bottom of the pool remains mostly unmelted due to low laser energy.

As the gap widens, laser energy begins reaching the bottom of the pool, causing the base of the keyhole to expand and the root to start melting through. This shifts the melt pool shape from a V-shape to a U-shape, increasing the angle between the sidewalls and the horizontal plane from β_1 to β_2 , as shown in Fig. 13b2. With a larger oscillation radius, the upper width of the molten pool expands, further decreasing β_1 , as shown in Fig. 13c1. When the gap remains is constant, a larger oscillation radius improves base material melting, allowing more liquid metal to reach the root. At an oscillation radius of 1.4 mm, the tangential angle (α) of the liquid-solid interface at the root also increases. Consequently, even though the melt pool's gravity increases as the oscillation radius increases from 0.8 mm to 1.4 mm, the support force and surface tension also rise, leading to greater weld gap tolerance.

Overall, the change in laser energy along the depth direction of the base material during O-LAHW is more sensitive to increasing gap size, leading to significant changes in molten pool shape. The sidewall angle β increases more in O-LAHW than in LAHW, as shown in Fig. 13a2,b2,c2. Moreover, the upper part of the molten pool absorbs more laser energy than the lower part during O-LAHW, reducing the melting at the root and decreasing the tangential angle α at the liquid-solid interface. As a result, the vertical surface tension in O-LAHW is lower compared to LAHW.

The mechanism of periodic collapse in O-LAHW is shown in Fig. 14. High-frequency laser beam oscillation divides the molten pool into an inner and outer ring (Fig. 14a,b). According to the force model discussed in subsection 3.4, the centrifugal effect of the high-frequency oscillations creates a virtual centrifugal force (P_c) on the melt pool. Using the beam oscillation ring as a dividing line, the virtual centrifugal pressure points from

the inner ring towards the outer ring. As previously discussed, when the gap increases to the point that the force equilibrium of the molten pool is disrupted, collapse first occurs in the middle of the molten pool (i.e., the inner ring), as shown in the high-speed photography results in Fig. 5c1. The outer ring, supported by the centrifugal effect, increases the support force between the molten pool and the side wall, delaying the collapse of the outer ring. According to Eq. 7, the larger the oscillation radius, the greater the P_c , which prolongs the duration of periodic collapse, as shown in Fig. 4b. The collapse of the inner-ring molten pool reduces the overall gravity of the pool. As seen in Fig. 5c, when the outer-ring molten pool collapses, the entire pool re-establishes equilibrium, which is then gradually disrupted again as the molten pool enlarges, leading to the periodic collapse process in O-LAHW, as depicted in Fig. 14c.



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Fig. 14. **Periodic collapse formation mechanism.** (a) Oscillating molten pool cross-section, green dotted lines and arrows represent the virtual centrifugal pressure (P_c), (b) top view of the oscillating molten pool, (c) periodic collapse welds;

Changes in the interaction between the laser and the base metal, caused by variations in the gap, significantly impact the molten pool's shape, which in turn influences the balance between surface tension, support force, gravity, and arc pressure. These dynamics can lead to pool collapse. Understanding the destabilization mechanism provides a strategy to enhance O-LAHW stability: by designing energy distribution to control the molten pool's shape, the forces acting on it can be adjusted to increase gap tolerance.

4. Conclusions

The study investigates the maximum gap tolerance of circular oscillating laser-arc hybrid welding (O-LAHW) and conventional laser-arc hybrid welding (LAHW), as well as the melt pool instability mechanism under large gaps. The main conclusions are as follows:

1. Increasing the oscillation radius from 0.8 mm to 1.4 mm improved gap tolerance from 1.39 mm to 1.68 mm. The oscillating molten pool exhibited a periodic collapse state before continuous collapse, thereby delaying the onset of continuous instability compared to non-oscillating conditions.
2. For the same gap, O-LAHW showed approximately a 60 % increase in gravity and a 10 % decrease in surface tension compared to LAHW. This leads to a higher tendency for melt pool destabilization and a lower gap tolerance after oscillation. The virtual centrifugal pressure generated by high-frequency oscillation is the primary driving force behind the periodic collapse of the melt pool.
3. Melt pool destabilization is not directly caused by changes in force magnitude but by changes in the melt pool's shape. Alterations in the angle of the slanting surface and the contact angle at the liquid-solid interface at the root reduce the vertical projections of the support force and surface tension, making it impossible to balance the melt pool's gravity and arc pressure, leading to destabilization.
4. Increasing the gap causes laser energy to deposit downward, reducing the contact angle at the liquid-solid interface at the root and, consequently, decreasing vertical upward

surface tension. However, increasing the oscillation diameter can mitigate this trend, thereby delaying the onset of collapse.

CRediT authorship contribution statement

Wei Liao: Conceptualization, Methodology, Data Curation, Writing - Original Draft, Writing - Review & Editing. **Benle Wang:** Methodology, Visualization, Investigation, Formal analysis. **Zhaoyang Wang:** Methodology, Data Curation. **Laihege Jiang:** Methodology, Software. **Ming Gao:** Conceptualization, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

References

- [1] A. Sahoo, S. Tripathy
Development in plasma arc welding process: a review
Mater. Today.: Proc., 41 (2021), pp. 363-368



[View PDF](#)

[View article](#)

[Google Scholar ↗](#)

- [2] S. Kilic, F. Ozturk, M.F. Demirdogen
A comprehensive literature review on friction stir welding: process parameters, joint integrity, and mechanical properties
J. Eng. Res. (2023)
[Google Scholar ↗](#)
- [3] S. Liu, Z. Wu, W. Zhou, H. Zhou, K. Zhang, D. Yin, Y. Lei, Y. Qiu
A review of welding simulation methods for large components
Prog. Nat. Sci.: Mater. Int., 33 (5) (2023), pp. 551-568
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [4] L. Guo, H. Wang, H. Liu, Y. Huang, Q. Wei, C.L. Leung, Y. Wu, H. Wang
Quantifying the effects of gap on the molten pool and porosity formation in laser butt welding
Int. J. Heat. Mass Transf., 209 (2023), Article 124143
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [5] H. Huang, P. Zhang, H. Yan, Z. Liu, Z. Yu, D. Wu, H. Shi, Y. Tian
Research on weld formation mechanism of laser-MIG arc hybrid welding with butt gap
Opt. Laser Technol., 133 (2021), Article 106530
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [6] Y.X. Zhang, S.W. Han, J. Cheon, S.J. Na, X. Gao
Effect of joint gap on bead formation in laser butt welding of stainless steel
J. Mater. Process. Technol., 249 (2017), pp. 274-284
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [7] J. Chen, X. Du, B. Kong, Y. Wei
Effects of groove clearance size on gap bridging capacity in PWLBW of Ti6Al4V alloy sheet assembled in butt joint configuration: numerical

simulation and experimental assessment

Opt. Laser Technol., 156 (2022), Article 108527



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [8] G. Turichin, I. Tsibulskiy, M. Kuznetsov, A. Akhmetov, M. Mildebrath, T. Hassel
Influence of the gap width on the geometry of the welded joint in hybrid laser-arc welding

Phys. Procedia, 78 (2015), pp. 14-23



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [9] J. Lamas, J. Frostevarg, A.F.H. Kaplan
Gap bridging for two modes of laser arc hybrid welding

J. Mater. Process. Technol., 224 (2015), pp. 73-79



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [10] J. Lamas, J. Karlsson, P. Norman, J. Powell, A.F. Kaplan, A. Yanez
The effect of fit-up geometry on melt flow and weld quality in laser hybrid welding

J. Laser Appl., 25 (3) (2013), Article 032010

[View in Scopus](#)

[Google Scholar](#)

- [11] Y.P. Kim, N. Alam, H.S. Bang, H.S. Bang
Observation of hybrid (cw Nd: YAG laser+ MIG) welding phenomenon in AA 5083 butt joints with different gap condition

Sci. Technol. Weld. Join., 11 (3) (2006), pp. 295-307

[View in Scopus](#)

[Google Scholar](#)

- [12] Z. Liu, Y. Fang, J. Qiu, M. Feng, Z. Luo, J. Yuan
Stabilization of weld pool through jet flow argon gas backing in C-Mn steel keyhole TIG welding

J. Mater. Process. Technol., 250 (2017), pp. 132-143



[View PDF](#)

[View article](#)

[View in Scopus](#)

[Google Scholar](#)

- [13] J. Frostevarg
Factors affecting weld root morphology in laser keyhole welding
Opt. Lasers Eng., 101 (2018), pp. 89-98
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [14] Z. Zhu, Z. Hu, H.L. Seet, T. Liu, W. Liao, U. Ramamurty, S.M. Nai
Recent progress on the additive manufacturing of aluminum alloys and aluminum matrix composites: microstructure, properties, and applications
Int. J. Mach. Tools Manuf., 190 (2023), Article 104047
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [15] R. Zhang, X. Tang, L. Xu, F. Lu, H. Cui
Mechanism study of thermal fluid flow and weld root hump suppression in full penetration laser welding of Al alloy with alternating magnetic field support
Int. J. Heat. Mass Transf., 166 (2021), Article 120759
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [16] M.E. Trybula, T. Gancarz, W. Gqsior
Density, surface tension and viscosity of liquid binary Al-Zn and ternary Al-Li-Zn alloys
Fluid phase equilibria, 421 (2016), pp. 39-48
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [17] L. Wang, Y. Liu, C. Yang, M. Gao
Study of porosity suppression in oscillating laser-MIG hybrid welding of AA6082 aluminum alloy
J. Mater. Process. Technol., 292 (2021), Article 117053
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [18] C. Cai, J. Xie, J. Yu, Y. Liu, J. Huang, H. Chen

Arc and keyhole behavior in narrow-gap oscillating laser-MIG hybrid welding of thick aluminum alloys

Opt. Laser Technol., 175 (2024 Aug 1), Article 110854



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[19]

Y. Meng, Q. Yu, M. Gao, H. Chen, Y. Jiang, Z. Yang

High-frequency oscillating laser-arc hybrid welding of 8-mm-thick high-strength aluminum alloy through synchronous wire-powder feeding

Int. J. Heat. Mass Transf., 222 (2024 May 1), Article 125180



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[20]

V. Schultz, T. Seefeld, F. Vollertsen

Gap bridging ability in laser beam welding of thin aluminum sheets

Phys. Procedia, 56 (2014), pp. 545-553



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[21]

H. Yu, Y. Yin, S. Ji, C. Zhang, T. Zhu, Z. Liu

Laser welding method and quality analysis of hairpin windings based on visual recognition of gap and displacement matching process

Opt. Laser Technol., 178 (2024), Article 111224



[View PDF](#)

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[View in Scopus ↗](#)

[Google Scholar ↗](#)

[22]

L. Wang, M. Gao, C. Zhang, X. Zeng

Effect of beam oscillating pattern on weld characterization of laser welding of AA6061-T6 aluminum alloy

Mater. Des., 108 (2016), pp. 707-717



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

[23]

P. Horník, H. Šebestová, J. Novotný, L. Mrňa

Laser beam oscillation strategy for weld geometry variation

J. Manuf. Process., 84 (2022), pp. 216-222



[View PDF](#)

[View article](#)

[View in Scopus ↗](#)

[Google Scholar ↗](#)

- [24] W. Liao, Y. Liu, M. Ma, D. Wang, L. Zhou, Z. Wang, M. Gong, L. Shi, M. Gao
Dynamic behaviors and homogenization mechanism of molten pool in laser-arc hybrid welding with beam oscillation
Opt. Laser Technol., 178 (2024), Article 111260
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [25] J. Han, Y. Shi, G. Zhang, K. Volodymyr, W.Y. Le
Minimizing defects and controlling the morphology of laser welded aluminum alloys using power modulation-based laser beam oscillation
J. Manuf. Process., 83 (2022), pp. 49-59
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [26] Q. Liu, D. Wu, Q. Wang, P. Zhang, H. Yan, T. Sun, J. Zeng, M. Yan, Z. Liu, R. Li
Research status of stability in dynamic process of LAHW based on droplet transfer behavior: a review
Coatings, 13 (1) (2023), p. 205
[Google Scholar ↗](#)
- [27] X. Li, S. Zhao, L. Jiang, Y. Lu, M. Gong, C. Zhang, M. Gao
Droplet detachment and motion behaviors in laser-MIG hybrid welding of aluminum alloy
Int. J. Heat. Mass Transf., 210 (2023), Article 124165
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [28] W.S. Chung, M. Hummel, C. Spurk, A. Häusler, A. Olowinsky, C. Häfner, F. Beckmann, J. Moosmann
In situ X-ray phase contrast imaging of the melt and vapor capillary behavior during the welding regime transition on aluminum with limited material thickness
Weld. World, 68 (1) (2024), pp. 43-50
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [29] P. Franciosa, A. Serino, R.A. Botros, D. Ceglarek
Closed-loop gap bridging control for remote laser welding of aluminum components based on first principle energy and mass balance
J. Laser Appl., 31 (2) (2019)
[Google Scholar ↗](#)
- [30] J.J. Lowke
Physical basis for the transition from globular to spray modes in gas metal arc welding
J. Phys. D: Appl. Phys., 42 (13) (2009), Article 135204
[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [31] S. Liu, F. Zhang, S. Dong, H. Zhang, F. Liu
Characteristics analysis of droplet transfer in laser-MAG hybrid welding process
Int. J. Heat. Mass Transf., 121 (2018), pp. 805-811
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [32] D. Wu, X. Hua, F. Li, L. Huang
Understanding of spatter formation in fiber laser welding of 5083 aluminum alloy
Int. J. Heat. Mass Transf., 113 (2017), pp. 730-740
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [33] T. Piatkowski, M. Wolski, T. Tomaszewski, P. Strzelecki, J. Sempruch
Analysis of the positioning process of objects on an oblique plane with barriers
Mech. Mach. Theory, 168 (2022), Article 104580
 [View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)
- [34] C. Wang, W. Suder, J. Ding, S. Williams

Wire based plasma arc and laser hybrid additive manufacture of Ti-6Al-4V

J. Mater. Process. Technol., 293 (2021), Article 117080



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [35] X. Meng, S.N. Putra, M. Bachmann, F. Yang, A. Artinov, M. Rethmeier
A fundamental study of physical mechanisms of wineglass-shaped fusion zone profile in laser melting

J. Mater. Process. Technol., 324 (2024), Article 118265



[View PDF](#) [View article](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [36] T. Patterson, J. Hochanadel, S. Sutton, B. Panton, J. Lippold
A review of high energy density beam processes for welding and additive manufacturing applications

Weld. World, 65 (7) (2021), pp. 1235-1306

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [37] A. Kaplan
A model of deep penetration laser welding based on calculation of the keyhole profile

J. Phys. D: Appl. Phys., 27 (9) (1994), p. 1805

[Crossref ↗](#) [View in Scopus ↗](#) [Google Scholar ↗](#)

- [38] X. Meng, S.N. Putra, M. Bachmann, M. Rethmeier
Influence of the spatial laser energy absorption on the molten pool dynamics in high-power laser beam welding

J. Laser Appl., 35 (4) (2023)

[Google Scholar ↗](#)

- [39] J. Chen, Z. Ouyang, X. Du, Y. Wei

Weld pool dynamics and joining mechanism in pulse wave laser beam welding of Ti-6Al-4V titanium alloy sheets assembled in butt joint with an air gap

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